

Building an Institutional-Grade Equity Research Pipeline

Data Management for Cross-Sectional Asset Pricing

B. Kofi Owusu Sarfo

Agenda

1

Dataset Overview

CRSP/Compustat via WRDS, 1.35M observations

2

Data Wrangling

Cleaning, feature engineering, 11 factors

3

Data Mining

Regression, classification, clustering, PCA

4

Visualizations

15 charts covering all required types

5

Portfolio & Risk

Markowitz optimization, Sharpe = 2.06

6

Conclusions

Key findings and implications

1. Dataset Overview

1,348,447

Firm-month
observations

14,666

Unique
securities

37

Raw
variables

34 yrs

Jan 1990 –
Dec 2023

CRSP Monthly Stock File: Returns, prices, shares outstanding for NYSE/AMEX/NASDAQ common stocks (shrcd 10,11)

Compustat Annual Fundamentals: Balance sheet and income statement (at, ceq, sale, ni, oibdp, capx)

Merge: CUSIP-8 backward asof join with 6-month availability lag (no look-ahead bias)

Source: WRDS (Wharton Research Data Services) - same infrastructure used by academic researchers and quantitative hedge funds

Data Quality & Ethics

Quality Checks

- ✓ Zero duplicate firm-months
- ✓ Zero null adjusted returns
- ✓ Zero non-positive market equity
- ✓ Share codes validated (10, 11)
- ✓ Exchange codes validated (1, 2, 3)

176 extreme returns → winsorized

58,538 negative B/E → flagged

Ethics Considerations

Survivorship bias: Delisting returns included

Data snooping: Temporal train/test split

Reproducibility: WRDS academic access

Privacy: Public financial data only

Market impact: Historical evidence, not guarantees

Exchange: NYSE 31.3% | AMEX 7.0% | NASDAQ 61.7% Missing: capx 8.2% (highest), most variables < 2%

2. Data Wrangling Pipeline

Step	Operation	Result
1–3	Parse dates, remove duplicates, drop null returns	1,348,447 rows (no removals)
4	Remove penny stocks ($ \text{price} < \1.00)	36,163 removed $\rightarrow$ 1,312,284
5	Winsorize returns (0.5th/99.5th pctl)	Extreme outliers clipped
6	Engineer 11 factors + 6 accounting ratios	37 $\rightarrow$ 65 columns
7	Winsorize ratios (1st/99th pctl per month)	Cross-sectional cleaning

Before

1,348,447 rows $\times$ 37 columns

Raw CRSP/Compustat merge output

After

1,312,284 rows $\times$ 65 columns

2.68% removed | 28 new features added

11 Engineered Cross-Sectional Factors

Factor	Formula	Coverage
Momentum	Cumulative ret t-12 to t-2	88.5%
Reversal	Prior month return	98.9%
Size	$\log(\text{lagged market equity})$	98.9%
Value (B/M)	$\text{book_equity} / \text{lagged ME}$	97.7%
Profitability	$\text{oibdp} / \text{total assets}$	97.7%
Investment	$\text{capx} / \text{total assets}$	91.7%
Leverage	$(\text{dltt} + \text{dlc}) / \text{total assets}$	98.7%
E/P	$\text{net income} / \text{lagged ME}$	97.5%
Beta	60-month rolling CAPM (mean=1.065)	67.9%
Volatility	12-month rolling std dev	94.6%
Size Quintile	NYSE breakpoints, all stocks assigned	—

3a. Regression - Return Prediction

Out-of-sample: Jan 2016 – Nov 2023 (216,744 obs) | Train: Jul 1993 – Dec 2015 (606,584 obs)

Model	OOS R ²	Hit Rate	L/S Spread	t-stat
Ridge	0.37%	53.76%	1.17%/mo	2.48
Lasso	0.58%	53.74%	1.26%/mo	2.56
Random Forest	0.54%	53.76%	1.44%/mo	3.88
XGBoost	0.30%	53.11%	1.43%/mo	4.38

Key insight: All models produce statistically significant long/short spreads ($t > 2.0$). Non-linear models (RF, XGBoost) achieve stronger economic results (1.43–1.44%/mo) vs. linear models (1.17–1.26%), suggesting factor interactions carry additional predictive information.

3b. Classification & Clustering

Classification (Up/Down)

Logistic Regression: 53.89% accuracy

RF Classifier: 53.97% accuracy

Both above 50% random baseline

Strong at predicting “up” (89% recall)

Weak at predicting “down” (16% recall)

K-Means Clustering (K=5)

5 firm archetypes identified:

C0 (17): Distressed turnaround firms

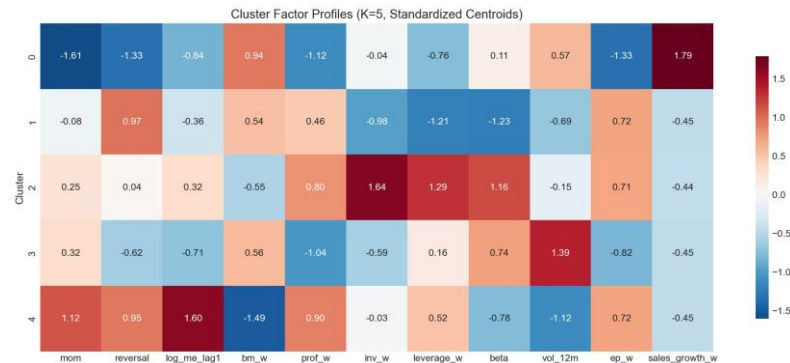
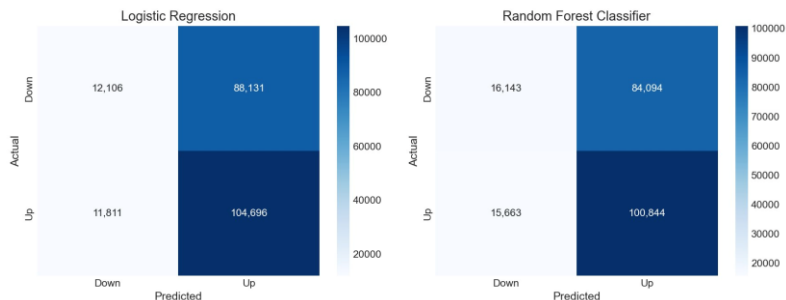
C1 (734): High-beta, high-leverage

C2 (523): Capital-intensive industrials

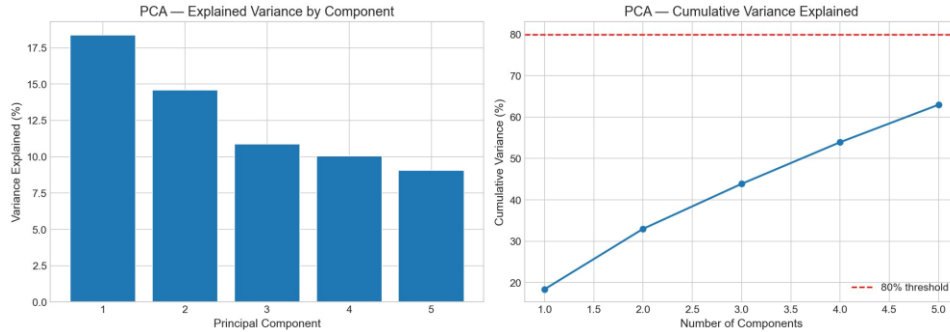
C3 (251): Unprofitable, speculative

C4 (914): Blue-chip growth (largest)

Classification — Confusion Matrices



3c. PCA & Correlation Analysis



Top Factor Correlations

Beta × Volatility: +0.385

Size × Value: -0.379

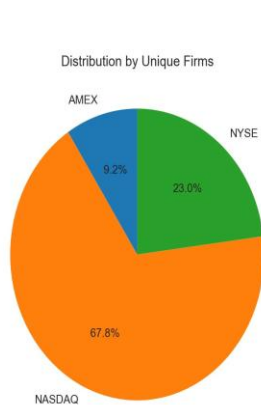
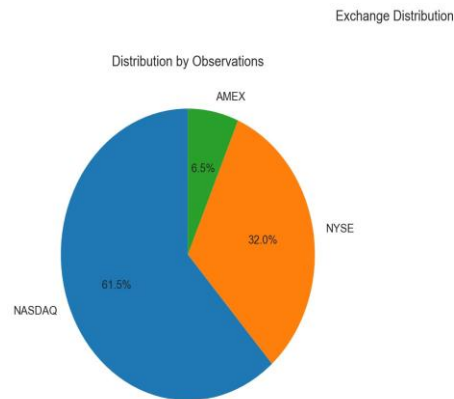
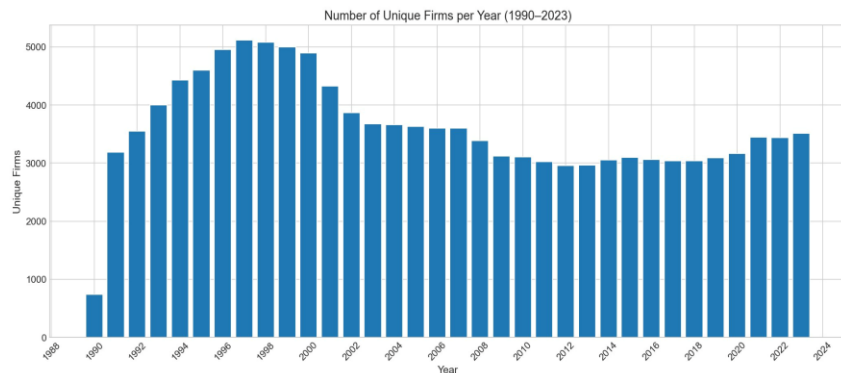
Profitability × E/P: +0.378

Max correlation < 0.4 →

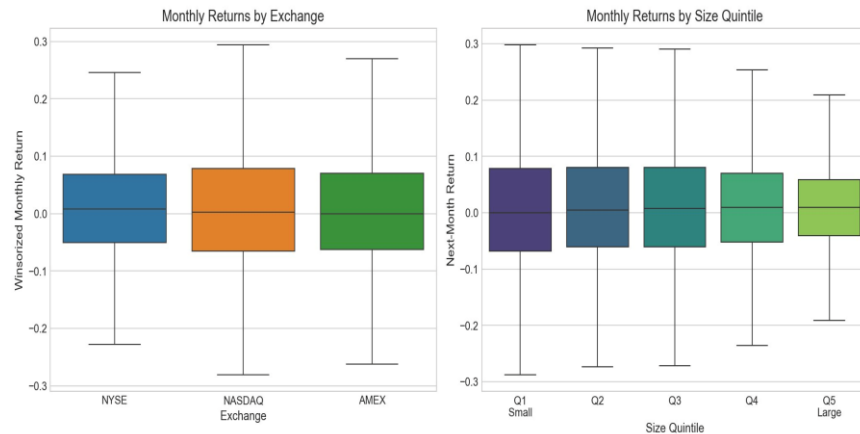
Factors are not redundant

PCA: 5 components capture 63% of variance. No single component dominates (max 18.4%), confirming 11 factors capture multiple independent risk dimensions. Text mining: not applicable — dataset contains only structured numerical/categorical data.

4. Data Visualizations



Box-Plot — Return Distributions



5. Portfolio Construction & Risk

2.06

Sharpe
Ratio

19.2%

Annual
Return

7.15%

Annual
Volatility

55

Stock
Positions

Top Holdings: SO (5.0%), DLTR (5.0%), CLX (4.9%), GIS (4.5%), EW (4.1%), MO (4.0%)

Portfolio tilts toward defensive/consumer staples (max-Sharpe favors steady returns) with high-momentum growth names (TSLA, NFLX).

Risk: 73.8% turnover vs equal-weight | 7.4 bps estimated transaction cost | Top 5 = 23.5% | Effective positions = 36

6. Conclusions

Factor prediction is real

All 4 models produce significant L/S spreads ($t = 2.48\text{--}4.38$) over 8-year OOS period

Non-linear models add value

RF/XGBoost achieve 1.43–1.44%/mo spread vs 1.17–1.26% for linear models

Multi-dimensional risk

PCA confirms 5 independent factor dimensions (63% variance) — no redundancy

Firm archetypes exist

K-Means identifies 5 interpretable clusters from blue-chip growth to distressed turnaround

Strong portfolio outcomes

Markowitz optimizer delivers Sharpe = 2.06, 19.2% annual return, 55-stock diversification

Thank You

Questions?

MIS 502 | Managing Data for Analytics
WPI Business School | Spring 2026